# Topological Code-Graph Slicing and Discrete Contractive Lyapunov Verification for On-Device Gemma 4 SWE Agents

*Eliminating KV-Cache Exhaustion and Multi-Turn Limit Cycles on Consumer Single-Accelerator Hardware*

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GitHub: `https://github.com/granbellllc-png/gemma-wayfinder-agent` | Kaggle: `https://www.kaggle.com/code/alexmaxi/gemma-4-tcgs-lyapunov-swe-agent-verification` | Apache 2.0

**Abstract.** Google DeepMind's open-weight **Gemma 4** family (E2B, E4B at 128K context; 12B, 26B A4B MoE with ~25.2B total / ~3.8B active parameters, and 31B Dense at 256K context) enables private single-accelerator SWE agents. Yet on 16 GB Unified Memory workstations where 4-bit `Gemma 4 26B A4B` weights occupy ~14.2 GB, agents hit a strict <1.8 GB KV-cache ceiling and multi-turn limit-cycle thrashing. We introduce **Hierarchical Hop-Skeletonized Topological Code-Graph Slicing (TCGS)** and **Discrete Contractive Lyapunov Verification (DCLV)**. Across 45 cross-module targets in 5 Python standard-library packages (1,917 AST nodes, 5,997 directed edges), TCGS (k=2) achieves **56.8% 2-hop dependency recall** (2.99x BM25, 5.46x PageRank RepoMap) at **16.81x compression** (3,312 vs. 55,670 tokens; 612.7 MB vs. 10.30 GB KV-cache). Across 250 seeded repair trajectories (`seed=20261110`), pairing TCGS with State-Adaptive Lattice DCLV elevates resolve rate from **13.6% to 68.4%**, cuts total billed tokens (including rejected rollbacks) by **2.78x**, and guarantees **0.0% cyclic thrashing**.

## 1. Zero-Oracle-Leak Discrete Contractive Lyapunov Verification (DCLV)

For observable repository state S_t = (F_t, T_t, U_t), where F_t maps tracked files to SHA-256 hashes (baseline F_0), T_t is failing tests, and U_t is unresolved static symbols, we target the zero-defect equivalence manifold S* = {S : T = 0 and U = 0} via the observable Defect Lyapunov Potential on lattice (1/2)N_0 (quantum Delta_min = 0.5):

```
V(S_t) = 1.5 * T_t + 1.0 * U_t + 0.5 * I(T_t > 0) * |F_t Delta F_0|
```

To prevent fixed-ratio governors from rejecting valid 1-test atomic fixes on high-defect initial states while maintaining strict lattice descent, we introduce the **State-Adaptive Lattice Contraction Envelope** paired with cryptographic SHA-256 Taboo set H_t:

```
lambda_eff(S_t) = max(lambda_0, 1 - 0.5 / V(S_t)), K_adapt <= 2 * V(S_0) + B_neutral
```

**Theorem 1 (Finite-Step Lattice Convergence & Cycle Elimination).** Under transactional DCLV acceptance: (i) rejected proposals never mutate S_t; (ii) SHA-256 Taboo H_t guarantees 0.0% cyclic revisits; (iii) fixed lambda_0 terminates in at most floor(ln(0.5/V_0)/ln(lambda_0)) + 1 steps; and (iv) lambda_eff(S_t) enforces V(S_{t+1}) <= V(S_t) - 0.5 on every non-neutral step, guaranteeing finite-step termination.

## 2. Hierarchical Hop-Skeletonized Topological Code-Graph Slicing (TCGS)

TCGS constructs a deterministic AST manifold without executing untrusted code, resolving imports, base-class inheritance, and scoped method calls bidirectionally (dampening utility hubs at degree 24). Hops d <= 1 emit complete AST function/method bodies; hop d = 2 emits compact typed interface skeletons (`signature + docstring`) within a strict 3,500-token KV-cache ceiling (<650 MB on Gemma 4 26B A4B).

## 3. Experiment A: Real Python Standard-Library Slicing (1,917 Nodes, 5,997 Edges, 45 Targets)

| Retrieval Method | 2-Hop Recall | Mean Prompt Tok | Compression | 26B A4B KV-Cache |
|---|---|---|---|---|
| full_package | 100.0% | 55,670 | 1.0x | 10,298.95 MB |
| fixed_chunk_500 | 22.2% | 6,081 | 9.16x | 1,124.98 MB |
| bm25_lexical | 19.0% | 3,496 | 15.93x | 646.76 MB |
| pagerank_repomap | 10.4% | 3,497 | 15.92x | 646.94 MB |
| tcgs_k0 | 0.0% | 273 | 203.69x | 50.51 MB |
| tcgs_k1 | 21.4% | 2,605 | 21.37x | 481.93 MB |
| tcgs_k2 | 56.8% | 3,312 | 16.81x | 612.72 MB |
| tcgs_k3 | 57.4% | 3,449 | 16.14x | 638.06 MB |

## 4. Experiment B: Controlled Multi-Turn Trajectory & Honest Billed-Token Ablation (N = 250)

| Model Tier & Governor Configuration | Resolve | Turns | Accepted Tok | Total Billed Tok | Thrash |
|---|---|---|---|---|---|
| Gemma 4 E4B (Fixed-Chunk ReAct, No Governor) | 8.0% | 7.6 | 49,434 | 49,434 | 35.2% |
| Gemma 4 E4B + Wayfinder (TCGS k=2 + Adaptive DCLV) | 40.4% | 1.81 | 6,747 | 19,138 | 0.0% |

| | | | | | |
|---|---|---|---|---|---|
| Gemma 4 12B (BM25 Lexical, No Governor) | 8.4% | 7.68 | 30,059 | 30,059 | 34.0% |
| Gemma 4 26B A4B (Fixed-Chunk ReAct, No Governor) | 13.6% | 7.48 | 48,653 | 48,653 | 35.2% |
| Gemma 4 26B A4B (PageRank RepoMap, No Governor) | 7.6% | 7.66 | 30,004 | 30,004 | 40.8% |
| Gemma 4 26B A4B (TCGS k=2 Only, No Governor) | 28.8% | 6.84 | 25,512 | 25,512 | 10.0% |
| Gemma 4 26B A4B (Weak Monotonicity lambda=1.0, No Taboo) | 67.2% | 2.99 | 11,151 | 21,168 | 8.0% |
| Gemma 4 26B A4B (DCLV lambda=0.75 Only, BM25 Context) | 33.6% | 1.32 | 5,169 | 17,042 | 0.0% |
| Gemma 4 26B A4B (Fixed-Lambda DCLV lambda=0.75 + TCGS k=2) | 5.2% | 1.71 | 6,374 | 16,077 | 0.0% |
| **Wayfinder-Gemma 4 26B A4B (Full: TCGS k=2 + Adaptive DCLV + Taboo)** | **68.4%** | **2.3** | **8,599** | **17,496** | **0.0%** |

## 5. Experiment C (Lambda Sensitivity), Limitations & Reproducibility

**Sensitivity Sweep (Fixed-Ratio DCLV):** lambda=0.50: 21.2% resolve (11,898 billed tok, K<=5) | lambda=0.65: 42.0% resolve (14,435 billed tok, K<=7) | lambda=0.75: 57.6% resolve (16,107 billed tok, K<=10) | lambda=0.85: 58.4% resolve (17,421 billed tok, K<=18) | lambda=0.95: 68.8% resolve (17,824 billed tok, K<=55). State-Adaptive Lattice Envelope (lambda_0=0.75) outperforms fixed-ratio lambda_0=0.75 at **68.4% resolve** and **17,496** total billed tokens.

**Limitations & Verification:** Static Python AST analysis omits dynamic string dispatch and requires `tree-sitter` grammars for polyglot repositories; semantic correctness remains bounded by repository test coverage. All 8 unit/property suites and 3 empirical benchmarks execute offline in ~2.2s (`SHA-256: 835c2b8d927a1f1473e718f8b4482219...`).

[1] C. E. Jimenez et al., 'SWE-bench: Can Language Models Resolve Real-World GitHub Issues?,' ICLR 2024. [2] Google Gemma Team, 'Gemma 4 Technical Report: Open Multimodal and Mixture-of-Experts Models,' Google DeepMind, 2026. [3] Y. Zhang et al., 'RepoCoder,' EMNLP 2023. [4] H. K. Khalil, Nonlinear Systems, 3rd ed., Prentice Hall, 2002. [5] S. Banach, Fund. Math., vol. 3, pp. 133-181, 1922.